Neural Computation -- 2015

92 article(s), all issues

1. The New Cosmos

David J. Eicher, Alex Filippenko

(no DOI)

No abstract available.

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2. Editorial

(authors unknown)

https://doi.org/10.1162/neco_e_00728 | pp. i-i

No abstract available.

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3. Neural Network Spectral Robustness under Perturbations of the Underlying Graph

Anca R■dulescu

https://doi.org/10.1162/neco_a_00798 | pp. 1-44

Recent studies have been using graph-theoretical approaches to model complex networks (such as social, infrastructural, or biological networks) and how their hardwired circuitry relates to their dynamic evolution in time. Understanding how configuration reflects on the coupled behavior in a system of dynamic nodes can be of great importance, for example, in the context of how the brain connectome is affecting brain function. However, the effect of connectivity patterns on network dynamics is far from being fully understood. We study the connections between edge configuration and dynamics in a simple oriented network composed of two interconnected cliques (representative of brain feedback regulatory circuitry). In this article our main goal is to study the spectra of the graph adjacency and Laplacian matrices, with a focus on three aspects in particular: (1) the sensitivity and robustness of the spectrum in response to varying the intra- and intermodular edge density, (2) the effects on the spectrum of perturbing the edge configuration while keeping the densities fixed, and (3) the effects of increasing the network size. We study some tractable aspects analytically, then simulate more general results numerically, thus aiming to motivate and explain our further work on the effect of these patterns on the network temporal dynamics and phase transitions. We discuss the implications of such results to modeling brain connectomics. We suggest potential applications to understanding synaptic restructuring in learning networks and the effects of network configuration on function of regulatory neural circuits.

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4. Sequential Tests for Large-Scale Learning

Anoop Korattikara, Yutian Chen, Max Welling

https://doi.org/10.1162/neco_a_00796 | pp. 45-70

We argue that when faced with big data sets, learning and inference algorithms should compute updates using only subsets of data items. We introduce algorithms that use sequential hypothesis tests to adaptively select such a subset of data points. The statistical properties of this subsampling process can be used to control the efficiency and accuracy of learning or inference. In the context of learning by optimization, we test for the probability that the update direction is no more than 90 degrees in the wrong direction. In the context of posterior inference using Markov chain Monte Carlo, we test for the probability that our decision to accept or reject a sample is wrong. We experimentally evaluate our algorithms on a number of models and data sets.

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5. A Note on Support Vector Machines with Polynomial Kernels

Hongzhi Tong

https://doi.org/10.1162/neco_a_00794 | pp. 71-88

We present a better theoretical foundation of support vector machines with polynomial kernels. The sample error is estimated under Tsybakov's noise assumption. In bounding the approximation error, we take advantage of a geometric noise assumption that was introduced to analyze gaussian kernels. Compared with the previous literature, the error analysis in this note does not require any regularity of the marginal distribution or smoothness of Bayes' rule. We thus establish the learning rates for polynomial kernels for a wide class of distributions.

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6. Analytical Calculation of Errors in Time and Value Perception Due to a Subjective Time Accumulator: A Mechanistic Model and the Generation of Weber’s Law

Vijay Mohan K Namboodiri, ■tefan Mihala■, Marshall G. Hussain Shuler

https://doi.org/10.1162/neco_a_00792 | pp. 89-117

It has been previously shown (Namboodiri, Mihalas, Marton, & Hussain Shuler, 2014) that an evolutionary theory of decision making and time perception is capable of explaining numerous behavioral observations regarding how humans and animals decide between differently delayed rewards of differing magnitudes and how they perceive time. An implementation of this theory using a stochastic drift-diffusion accumulator model (Namboodiri, Mihalas, & Hussain Shuler, 2014a) showed that errors in time perception and decision making approximately obey Weber's law for a range of parameters. However, prior calculations did not have a clear mechanistic underpinning. Further, these calculations were only approximate, with the range of parameters being limited. In this letter, we provide a full analytical treatment of such an accumulator model, along with a mechanistic implementation, to calculate the expression of these errors for the entirety of the parameter space. In our mechanistic model, Weber's law results from synaptic facilitation and depression within the feedback synapses of the accumulator. Our theory also makes the prediction that the steepness of temporal discounting can be affected by requiring the precise timing of temporal intervals. Thus, by presenting exact quantitative calculations, this work provides falsifiable predictions for future experimental testing.

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7. Efficient Associative Computation with Discrete Synapses

Andreas Knoblauch

https://doi.org/10.1162/neco_a_00795 | pp. 118-186

Neural associative networks are a promising computational paradigm for both modeling neural circuits of the brain and implementing associative memory and Hebbian cell assemblies in parallel VLSI or nanoscale hardware. Previous work has extensively investigated synaptic learning in linear models of the Hopfield type and simple nonlinear models of the Steinbuch/Willshaw type. Optimized Hopfield networks of size n can store a large number of about n(2)/k memories of size k (or associations between them) but require real-valued synapses, which are expensive to implement and can store at most C = 0.72 bits per synapse. Willshaw networks can store a much smaller number of about n(2)/k(2) memories but get along with much cheaper binary synapses. Here I present a learning model employing synapses with discrete synaptic weights. For optimal discretization parameters, this model can store, up to a factor ζ close to one, the same number of memories as for optimized Hopfield-type learning--for example, ζ = 0.64 for binary synapses, ζ = 0.88 for 2 bit (four-state) synapses, ζ = 0.96 for 3 bit (8-state) synapses, and ζ > 0.99 for 4 bit (16-state) synapses. The model also provides the theoretical framework to determine optimal discretization parameters for computer implementations or brainlike parallel hardware including structural plasticity. In particular, as recently shown for the Willshaw network, it is possible to store C(I) = 1 bit per computer bit and up to C(S) = log n bits per nonsilent synapse, whereas the absolute number of stored memories can be much larger than for the Willshaw model.

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8. Reduction of Trial-to-Trial Perceptual Variability by Intracortical Tonic Inhibition

Osamu Hoshino, Meihong Zheng, Kazuo Watanabe

https://doi.org/10.1162/neco_a_00799 | pp. 187-215

Variability is a prominent characteristic of cognitive brain function. For instance, different trials of presentation of the same stimulus yield higher variability in its perception: subjects sometimes fail in perceiving the same stimulus. Perceptual variability could be attributable to ongoing-spontaneous fluctuation in neuronal activity prior to sensory stimulation. Simulating a cortical neural network model, we investigated the underlying neuronal mechanism of perceptual variability in relation to variability in ongoing-spontaneous neuronal activity. In the network model, populations of principal cells (cell assemblies) encode information about sensory features. Each cell assembly is sensitive to one particular feature stimulus. Transporters on GABAergic interneurons regulate ambient GABA concentration in a neuronal activity-dependent manner. Ambient GABA molecules activate extrasynaptic GABAa receptors on principal cells and interneurons, and provide them with tonic inhibitory currents. We controlled the variability of ongoing-spontaneous neuronal activity by manipulating the basal level of ambient GABA and assessed the perceptual performance of the network: detection of a feature stimulus. In an erroneous response, stimulus-irrelevant but not stimulus-relevant principal cells were activated, generating trains of action potentials. Perceptual variability, reflected in error rate in detecting the same stimulus that was presented repeatedly to the network, was increased as the variability in ongoing-spontaneous membrane potential among cell assemblies increased. Frequent, transient membrane depolarization below firing threshold was the major cause of the increased neuronal variability, for which a decrease in basal ambient GABA concentration was responsible. We suggest that ambient GABA in the brain may have a role in reducing the variability in ongoing-spontaneous

neuronal activity, leading to a decrease in perceptual variability and therefore to reliable sensory perception.

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9. An Empirical Overview of the No Free Lunch Theorem and Its Effect on Real-World Machine Learning Classification

David Gómez-Guillén, Alfonso Rojas■Domínguez

https://doi.org/10.1162/neco_a_00793 | pp. 216-228

A sizable amount of research has been done to improve the mechanisms for knowledge extraction such as machine learning classification or regression. Quite unintuitively, the no free lunch (NFL) theorem states that all optimization problem strategies perform equally well when averaged over all possible problems. This fact seems to clash with the effort put forth toward better algorithms. This letter explores empirically the effect of the NFL theorem on some popular machine learning classification techniques over real-world data sets.

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10. Global Exponential Stability of Cohen-Grossberg Neural Networks with Piecewise Constant Argument of Generalized Type and Impulses

Qiang Xi

https://doi.org/10.1162/neco_a_00797 | pp. 229-255

In this letter, we consider a model of Cohen-Grossberg neural networks with piecewise constant argument of generalized type and impulses. Sufficient conditions ensuring the existence and uniqueness of solutions are obtained. Based on constructing a new differential inequality with piecewise constant argument and impulse and using the Lyapunov function method, we derive sufficient conditions ensuring the global exponential stability of equilibrium point, with approximate exponential convergence rate. An example is given to illustrate the validity and advantage of the theoretical results.

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11. Correlational Neural Networks

Sarath Chandar, Mitesh M. Khapra, Hugo Larochelle, Balaraman Ravindran

https://doi.org/10.1162/neco_a_00801 | pp. 257-285

Common representation learning (CRL), wherein different descriptions (or views) of the data are embedded in a common subspace, has been receiving a lot of attention recently. Two popular paradigms here are canonical correlation analysis (CCA)-based approaches and autoencoder (AE)-based approaches. CCA-based approaches learn a joint representation by maximizing correlation of the views when projected to the common subspace. AE-based methods learn a common representation by minimizing the error of reconstructing the two views. Each of these approaches has its own advantages and disadvantages. For example, while CCA-based approaches outperform AE-based approaches for the task of transfer learning, they are not as scalable as the latter. In this work, we propose an AE-based approach, correlational neural network (CorrNet), that explicitly maximizes correlation among the views when projected to the common subspace. Through a series of experiments, we demonstrate that the proposed CorrNet is better than AE and CCA with respect to its ability to learn correlated common representations. We employ CorrNet for several cross-language tasks and show that the representations learned using it perform better than the ones learned using other state-of-the-art approaches.

12. Timing of Cortical Events Preceding Voluntary Movement

Victor Vvedensky, Andrey O. Prokofyev

https://doi.org/10.1162/neco_a_00802 | pp. 286-304

We studied magnetic signals from the human brain recorded during a second before a self-paced finger movement. Sharp triangular peaks were observed in the averaged signals about 0.7 second before the finger movement. The amplitude of the peaks varied considerably from trial to trial, which indicated that the peaks were concurrent with much longer oscillatory processes. One can cluster trials into distinct groups with characteristic sequences of events. Prominent short trains of pulses in the beta frequency band were identified in the premovement period. This observation suggests that during preparation of the intended movement, cortical activity is well organized in time but differs from trial to trial. Magnetoencephalography can capture these processes with high temporal resolution and allows their study in fine detail.

13. Mutual Information, Fisher Information, and Efficient Coding

Xue-Xin Wei, Alan A. Stocker

https://doi.org/10.1162/neco_a_00804 | pp. 305-326

Fisher information is generally believed to represent a lower bound on mutual information (Brunel & Nadal, 1998), a result that is frequently used in the assessment of neural coding efficiency. However, we demonstrate that the relation between these two quantities is more nuanced than previously thought. For example, we find that in the small noise regime, Fisher information actually provides an upper bound on mutual information. Generally our results show that it is more appropriate to consider Fisher information as an approximation rather than a bound on mutual information. We analytically derive the correspondence between the two quantities and the conditions under which the approximation is good. Our results have implications for neural coding theories and the link between neural population coding and psychophysically measurable behavior. Specifically, they allow us to formulate the efficient coding problem of maximizing mutual information between a stimulus variable and the response of a neural population in terms of Fisher information. We derive a signature of efficient coding expressed as the correspondence between the population Fisher information and the distribution of the stimulus variable. The signature is more general than previously proposed solutions that rely on specific assumptions about the neural tuning characteristics. We demonstrate that it can explain measured tuning characteristics of cortical neural populations that do not agree with previous models of efficient coding.

14. Memory Stacking in Hierarchical Networks

Johan Westö, Patrick May, Hannu Tiitinen

https://doi.org/10.1162/neco_a_00803 | pp. 327-353

Robust representations of sounds with a complex spectrotemporal structure are thought to emerge in hierarchically organized auditory cortex, but the computational advantage of this hierarchy remains unknown. Here, we used computational models to study how such hierarchical structures affect temporal binding in neural networks. We equipped individual units in different types of feedforward networks with local memory mechanisms storing recent inputs and observed how this affected the ability of the networks to

process stimuli context dependently. Our findings illustrate that these local memories stack up in hierarchical structures and hence allow network units to exhibit selectivity to spectral sequences longer than the time spans of the local memories. We also illustrate that short-term synaptic plasticity is a potential local memory mechanism within the auditory cortex, and we show that it can bring robustness to context dependence against variation in the temporal rate of stimuli, while introducing nonlinearities to response profiles that are not well captured by standard linear spectrotemporal receptive field models. The results therefore indicate that short-term synaptic plasticity might provide hierarchically structured auditory cortex with computational capabilities important for robust representations of spectrotemporal patterns.

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15. Infinite Continuous Feature Model for Psychiatric Comorbidity Analysis

Isabel Valera, Francisco J. R. Ruiz, Pablo M. Olmos, Carlos Blanco, Fernando Pérez■Cruz

https://doi.org/10.1162/neco_a_00805 | pp. 354-381

We aim at finding the comorbidity patterns of substance abuse, mood and personality disorders using the diagnoses from the National Epidemiologic Survey on Alcohol and Related Conditions database. To this end, we propose a novel Bayesian nonparametric latent feature model for categorical observations, based on the Indian buffet process, in which the latent variables can take values between 0 and 1. The proposed model has several interesting features for modeling psychiatric disorders. First, the latent features might be off, which allows distinguishing between the subjects who suffer a condition and those who do not. Second, the active latent features take positive values, which allows modeling the extent to which the patient has that condition. We also develop a new Markov chain Monte Carlo inference algorithm for our model that makes use of a nested expectation propagation procedure.

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16. Filtering with State-Observation Examples via Kernel Monte Carlo Filter

Motonobu Kanagawa, Yu Nishiyama, Arthur Gretton, Kenji Fukumizu

https://doi.org/10.1162/neco_a_00806 | pp. 382-444

This letter addresses the problem of filtering with a state-space model. Standard approaches for filtering assume that a probabilistic model for observations (i.e., the observation model) is given explicitly or at least parametrically. We consider a setting where this assumption is not satisfied; we assume that the knowledge of the observation model is provided only by examples of state-observation pairs. This setting is important and appears when state variables are defined as quantities that are very different from the observations. We propose kernel Monte Carlo filter, a novel filtering method that is focused on this setting. Our approach is based on the framework of kernel mean embeddings, which enables nonparametric posterior inference using the state-observation examples. The proposed method represents state distributions as weighted samples, propagates these samples by sampling, estimates the state posteriors by kernel Bayes' rule, and resamples by kernel herding. In particular, the sampling and resampling procedures are novel in being expressed using kernel mean embeddings, so we theoretically analyze their behaviors. We reveal the following properties, which are similar to those of corresponding procedures in particle methods: the performance of sampling can degrade if the effective sample size of a weighted sample is small, and resampling improves the sampling performance by increasing the effective sample size. We first demonstrate these theoretical findings by synthetic experiments. Then we show the effectiveness of the proposed filter by artificial and real data experiments, which include vision-based mobile robot localization.

17. Spontaneous Motion on Two-Dimensional Continuous Attractors

Chi Chung Alan Fung, ■■■-■■■ ■■■■■

https://doi.org/10.1162/neco_a_00711 | pp. 507-547

Attractor models are simplified models used to describe the dynamics of firing rate profiles of a pool of neurons. The firing rate profile, or the neuronal activity, is thought to carry information. Continuous attractor neural networks (CANNs) describe the neural processing of continuous information such as object position, object orientation, and direction of object motion. Recently it was found that in one-dimensional CANNs, short-term synaptic depression can destabilize bump-shaped neuronal attractor activity profiles. In this article, we study two-dimensional CANNs with short-term synaptic depression and spike frequency adaptation. We found that the dynamics of CANNs with short-term synaptic depression and CANNs with spike frequency adaptation are qualitatively similar. We also found that in both kinds of CANNs, the perturbative approach can be used to predict phase diagrams, dynamical variables, and speed of spontaneous motion.

18. Oscillator-Interference Models of Path Integration Do Not Require Theta Oscillations

Jeff Orchard

https://doi.org/10.1162/neco_a_00701 | pp. 548-560

Navigation and path integration in rodents seems to involve place cells, grid cells, and theta oscillations (4-12 Hz) in the local field potential. Two main theories have been proposed to explain the neurological underpinnings of how these phenomena relate to navigation and to each other. Attractor network (AN) models revolve around the idea that local excitation and long-range inhibition connectivity can spontaneously generate grid-cell-like activity patterns. Oscillator interference (OI) models propose that spatial patterns of activity are caused by the interference patterns between neural oscillators. In rats, these oscillators have a frequency close to the theta frequency. Recent studies have shown that bats do not exhibit a theta cycle when they crawl, and yet they still have grid cells. This has been interpreted as a criticism of OI models. However, OI models do not require theta oscillations. We explain why the absence of theta oscillations does not contradict OI models and discuss how the two families of models might be distinguished experimentally.

19. What Can Neuromorphic Event-Driven Precise Timing Add to Spike-Based Pattern Recognition?

Himanshu Akolkar, Cédric Meyer, Xavier Clady, Olivier Marre, Chiara Bartolozzi, Stefano Panzeri, Ryad Benosman

https://doi.org/10.1162/neco_a_00703 | pp. 561-593

This letter introduces a study to precisely measure what an increase in spike timing precision can add to spike-driven pattern recognition algorithms. The concept of generating spikes from images by converting gray levels into spike timings is currently at the basis of almost every spike-based modeling of biological visual systems. The use of images naturally leads to generating incorrect artificial and redundant spike

timings and, more important, also contradicts biological findings indicating that visual processing is massively parallel, asynchronous with high temporal resolution. A new concept for acquiring visual information through pixel-individual asynchronous level-crossing sampling has been proposed in a recent generation of asynchronous neuromorphic visual sensors. Unlike conventional cameras, these sensors acquire data not at fixed points in time for the entire array but at fixed amplitude changes of their input, resulting optimally sparse in space and time-pixel individually and precisely timed only if new, (previously unknown) information is available (event based). This letter uses the high temporal resolution spiking output of neuromorphic event-based visual sensors to show that lowering time precision degrades performance on several recognition tasks specifically when reaching the conventional range of machine vision acquisition frequencies (30-60 Hz). The use of information theory to characterize separability between classes for each temporal resolution shows that high temporal acquisition provides up to 70% more information that conventional spikes generated from frame-based acquisition as used in standard artificial vision, thus drastically increasing the separability between classes of objects. Experiments on real data show that the amount of information loss is correlated with temporal precision. Our information-theoretic study highlights the potentials of neuromorphic asynchronous visual sensors for both practical applications and theoretical investigations. Moreover, it suggests that representing visual information as a precise sequence of spike times as reported in the retina offers considerable advantages for neuro-inspired visual computations.

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20. Delay Differential Analysis of Time Series

Claudia Lainscsek, Terrence J. Sejnowski

https://doi.org/10.1162/neco_a_00706 | pp. 594-614

Nonlinear dynamical system analysis based on embedding theory has been used for modeling and prediction, but it also has applications to signal detection and classification of time series. An embedding creates a multidimensional geometrical object from a single time series. Traditionally either delay or derivative embeddings have been used. The delay embedding is composed of delayed versions of the signal, and the derivative embedding is composed of successive derivatives of the signal. The delay embedding has been extended to nonuniform embeddings to take multiple timescales into account. Both embeddings provide information on the underlying dynamical system without having direct access to all the system variables. Delay differential analysis is based on functional embeddings, a combination of the derivative embedding with nonuniform delay embeddings. Small delay differential equation (DDE) models that best represent relevant dynamic features of time series data are selected from a pool of candidate models for detection or classification. We show that the properties of DDEs support spectral analysis in the time domain where nonlinear correlation functions are used to detect frequencies, frequency and phase couplings, and bispectra. These can be efficiently computed with short time windows and are robust to noise. For frequency analysis, this framework is a multivariate extension of discrete Fourier transform (DFT), and for higher-order spectra, it is a linear and multivariate alternative to multidimensional fast Fourier transform of multidimensional correlations. This method can be applied to short or sparse time series and can be extended to cross-trial and cross-channel spectra if multiple short data segments of the same experiment are available. Together, this time-domain toolbox provides higher temporal resolution, increased frequency and phase coupling information, and it allows an easy and straightforward implementation of higher-order spectra across time compared with frequency-based methods such as the DFT and cross-spectral analysis.

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21. Range-Based ICA Using a Nonsmooth Quasi-Newton Optimizer for Electroencephalographic Source Localization in Focal Epilepsy

S. Easter Selvan, S. Thomas George, R. Balakrishnan

https://doi.org/10.1162/neco_a_00700 | pp. 628-671

Independent component analysis (ICA) aims at separating a multivariate signal into independent nongaussian signals by optimizing a contrast function with no knowledge on the mixing mechanism. Despite the availability of a constellation of contrast functions, a Hartley-entropy-based ICA contrast endowed with the discriminacy property makes it an appealing choice as it guarantees the absence of mixing local optima. Fueled by an outstanding source separation performance of this contrast function in practical instances, a succession of optimization techniques has recently been adopted to solve the ICA problem. Nevertheless, the nondifferentiability of the considered contrast restricts the choice of the optimizer to the class of derivative-free methods. On the contrary, this letter concerns a Riemannian quasi-Newton scheme involving an explicit expression for the gradient to optimize the contrast function that is differentiable almost everywhere. Furthermore, the inexact line search insisting on the weak Wolfe condition and a terminating criterion befitting the partly smooth function optimization have been generalized to manifold settings, leaving the previous results intact. The investigations with diversified images and the electroencephalographic (EEG) data acquired from 45 focal epileptic subjects demonstrate the efficacy of our approach in terms of computational savings and reliable EEG source localization, respectively. Additional experimental results are available in the online supplement.

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22. Two-Trace Model for Spike-Timing-Dependent Synaptic Plasticity

Rodrigo Echeveste, Claudius Gros

https://doi.org/10.1162/neco_a_00707 | pp. 672-698

We present an effective model for timing-dependent synaptic plasticity (STDP) in terms of two interacting traces, corresponding to the fraction of activated NMDA receptors and the [Formula: see text] concentration in the dendritic spine of the postsynaptic neuron. This model intends to bridge the worlds of existing simplistic phenomenological rules and highly detailed models, thus constituting a practical tool for the study of the interplay of neural activity and synaptic plasticity in extended spiking neural networks. For isolated pairs of pre- and postsynaptic spikes, the standard pairwise STDP rule is reproduced, with appropriate parameters determining the respective weights and timescales for the causal and the anticausal contributions. The model contains otherwise only three free parameters, which can be adjusted to reproduce triplet nonlinearities in hippocampal culture and cortical slices. We also investigate the transition from time-dependent to rate-dependent plasticity occurring for both correlated and uncorrelated spike patterns.

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23. On Firing Rate Estimation for Dependent Interspike Intervals

Elisa Benedetto, Federico Polito, Laura Sacerdote

https://doi.org/10.1162/neco_a_00709 | pp. 699-724

If interspike intervals are dependent, the instantaneous firing rate does not catch important features of spike trains. In this case, the conditional instantaneous rate plays the role of the instantaneous firing rate for the case of samples of independent interspike intervals. If the conditional distribution of the interspikes intervals

(ISIs) is unknown, it becomes difficult to evaluate the conditional firing rate. We propose a nonparametric estimator for the conditional instantaneous firing rate for Markov, stationary, and ergodic ISIs. An algorithm to check the reliability of the proposed estimator is introduced, and its consistency properties are proved. The method is applied to data obtained from a stochastic two-compartment model and to in vitro experimental data.

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24. Memristor Models for Machine Learning

Juan Pablo Carbajal, Joni Dambre, Michiel Hermans, Benjamin Schrauwen

https://doi.org/10.1162/neco_a_00694 | pp. 725-747

In the quest for alternatives to traditional complementary metal-oxide-semiconductor, it is being suggested that digital computing efficiency and power can be improved by matching the precision to the application. Many applications do not need the high precision that is being used today. In particular, large gains in area and power efficiency could be achieved by dedicated analog realizations of approximate computing engines. In this work we explore the use of memristor networks for analog approximate computation, based on a machine learning framework called reservoir computing. Most experimental investigations on the dynamics of memristors focus on their nonvolatile behavior. Hence, the volatility that is present in the developed technologies is usually unwanted and is not included in simulation models. In contrast, in reservoir computing, volatility is not only desirable but necessary. Therefore, in this work, we propose two different ways to incorporate it into memristor simulation models. The first is an extension of Strukov's model, and the second is an equivalent Wiener model approximation. We analyze and compare the dynamical properties of these models and discuss their implications for the memory and the nonlinear processing capacity of memristor networks. Our results indicate that device variability, increasingly causing problems in traditional computer design, is an asset in the context of reservoir computing. We conclude that although both models could lead to useful memristor-based reservoir computing systems, their computational performance will differ. Therefore, experimental modeling research is required for the development of accurate volatile memristor models.

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25. On Resolving Simultaneous Congruences Using Belief Propagation

Yongseok Yoo, Sriram Vishwanath

https://doi.org/10.1162/neco_a_00702 | pp. 748-770

Graphical models and related algorithmic tools such as belief propagation have proven to be useful tools in (approximately) solving combinatorial optimization problems across many application domains. A particularly combinatorially challenging problem is that of determining solutions to a set of simultaneous congruences. Specifically, a continuous source is encoded into multiple residues with respect to distinct moduli, and the goal is to recover the source efficiently from noisy measurements of these residues. This problem is of interest in multiple disciplines, including neural codes, decentralized compression in sensor networks, and distributed consensus in information and social networks. This letter reformulates the recovery problem as an optimization over binary latent variables. Then we present a belief propagation algorithm, a layered variant of affinity propagation, to solve the problem. The underlying encoding structure of multiple congruences naturally results in a layered graphical model for the problem, over which the algorithms are deployed, resulting in a layered affinity propagation (LAP) solution. First, the convergence of

LAP to an approximation of the maximum likelihood (ML) estimate is shown. Second, numerical simulations show that LAP converges within a few iterations and that the mean square error of LAP approaches that of the ML estimation at high signal-to-noise ratios.

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26. Structural Intervention Distance for Evaluating Causal Graphs

Jonas Peters, Peter Bühlmann

https://doi.org/10.1162/neco_a_00708 | pp. 771-799

Causal inference relies on the structure of a graph, often a directed acyclic graph (DAG). Different graphs may result in different causal inference statements and different intervention distributions. To quantify such differences, we propose a (pre-)metric between DAGs, the structural intervention distance (SID). The SID is based on a graphical criterion only and quantifies the closeness between two DAGs in terms of their corresponding causal inference statements. It is therefore well suited for evaluating graphs that are used for computing interventions. Instead of DAGs, it is also possible to compare CPDAGs, completed partially DAGs that represent Markov equivalence classes. The SID differs significantly from the widely used structural Hamming distance and therefore constitutes a valuable additional measure. We discuss properties of this distance and provide a (reasonably) efficient implementation with software code available on the first author's home page.

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27. Spontaneous Action Potentials and Neural Coding in Unmyelinated Axons

Cian O'Donnell, Mark C. W. van Rossum

https://doi.org/10.1162/neco_a_00705 | pp. 801-818

The voltage-gated Na and K channels in neurons are responsible for action potential generation. Because ion channels open and close in a stochastic fashion, spontaneous (ectopic) action potentials can result even in the absence of stimulation. While spontaneous action potentials have been studied in detail in single-compartment models, studies on spatially extended processes have been limited. The simulations and analysis presented here show that spontaneous rate in unmyelinated axon depends nonmonotonically on the length of the axon, that the spontaneous activity has sub-Poisson statistics, and that neural coding can be hampered by the spontaneous spikes by reducing the probability of transmitting the first spike in a train.

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28. Accurate Connection Strength Estimation Based on Variational Bayes for Detecting Synaptic Plasticity

Takuya Isomura, Yutaro Ogawa, Kiyoshi Kotani, Yasuhiko Jimbo

https://doi.org/10.1162/neco_a_00721 | pp. 819-844

Connection strength estimation is widely used in detecting the topology of neuronal networks and assessing their synaptic plasticity. A recently proposed model-based method using the leaky integrate-and-fire model neuron estimates membrane potential from spike trains by calculating the maximum a posteriori (MAP) path. We further enhance the MAP path method using variational Bayes and dynamic causal modeling. Several

simulations demonstrate that the proposed method can accurately estimate connection strengths with an error ratio of less than 20%. The results suggest that the proposed method can be an effective tool for detecting network structure and synaptic plasticity.

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29. Hardware-Amenable Structural Learning for Spike-Based Pattern Classification Using a Simple Model of Active Dendrites

Shaista Hussain, Shih■Chii Liu, Arindam Basu

https://doi.org/10.1162/neco_a_00713 | pp. 845-897

This letter presents a spike-based model that employs neurons with functionally distinct dendritic compartments for classifying high-dimensional binary patterns. The synaptic inputs arriving on each dendritic subunit are nonlinearly processed before being linearly integrated at the soma, giving the neuron the capacity to perform a large number of input-output mappings. The model uses sparse synaptic connectivity, where each synapse takes a binary value. The optimal connection pattern of a neuron is learned by using a simple hardware-friendly, margin-enhancing learning algorithm inspired by the mechanism of structural plasticity in biological neurons. The learning algorithm groups correlated synaptic inputs on the same dendritic branch. Since the learning results in modified connection patterns, it can be incorporated into current event-based neuromorphic systems with little overhead. This work also presents a branch-specific spike-based version of this structural plasticity rule. The proposed model is evaluated on benchmark binary classification problems, and its performance is compared against that achieved using support vector machine and extreme learning machine techniques. Our proposed method attains comparable performance while using 10% to 50% less in computational resource than the other reported techniques.

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30. Neuronal Calcium Wave Propagation Varies with Changes in Endoplasmic Reticulum Parameters: A Computer Model

Samuel A. Neymotin, Robert A. McDougal, Mohamed Sherif, Christopher P. Fall, Michael L. Hines, William W. Lytton

https://doi.org/10.1162/neco_a_00712 | pp. 898-924

Calcium ([Formula: see text]) waves provide a complement to neuronal electrical signaling, forming a key part of a neuron’s second messenger system. We developed a reaction-diffusion model of an apical dendrite with diffusible inositol triphosphate ([Formula: see text]), diffusible [Formula: see text], [Formula: see text] receptors ([Formula: see text]s), endoplasmic reticulum (ER) [Formula: see text] leak, and ER pump (SERCA) on ER. [Formula: see text] is released from ER stores via [Formula: see text]s upon binding of [Formula: see text] and [Formula: see text]. This results in [Formula: see text]-induced-[Formula: see text]-release (CICR) and increases [Formula: see text] spread. At least two modes of [Formula: see text] wave spread have been suggested: a continuous mode based on presumed relative homogeneity of ER within the cell and a pseudo-saltatory model where [Formula: see text] regeneration occurs at discrete points with diffusion between them. We compared the effects of three patterns of hypothesized [Formula: see text] distribution: (1) continuous homogeneous ER, (2) hotspots with increased [Formula: see text] density ([Formula: see text] hotspots), and (3) areas of increased ER density (ER stacks). All three modes produced [Formula: see text] waves with velocities similar to those measured in vitro (approximately 50–90 [Formula:

see text]m /sec). Continuous ER showed high sensitivity to [Formula: see text] density increases, with time to onset reduced and speed increased. Increases in SERCA density resulted in opposite effects. The measures were sensitive to changes in density and spacing of [Formula: see text] hotspots and stacks. Increasing the apparent diffusion coefficient of [Formula: see text] substantially increased wave speed. An extended electrochemical model, including voltage-gated calcium channels and AMPA synapses, demonstrated that membrane priming via AMPA stimulation enhances subsequent [Formula: see text] wave amplitude and duration. Our modeling suggests that pharmacological targeting of [Formula: see text]s and SERCA could allow modulation of [Formula: see text] wave propagation in diseases where [Formula: see text] dysregulation has been implicated.

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31. Visual Tracking Using Neuromorphic Asynchronous Event-Based Cameras

Zhenjiang Ni, Sio-Hoï Ieng, C. Posch, Jean-Claude Régnier, Ryad Benosman
https://doi.org/10.1162/neco_a_00720 | pp. 925-953

This letter presents a novel computationally efficient and robust pattern tracking method based on a time-encoded, frame-free visual data. Recent interdisciplinary developments, combining inputs from engineering and biology, have yielded a novel type of camera that encodes visual information into a continuous stream of asynchronous, temporal events. These events encode temporal contrast and intensity locally in space and time. We show that the sparse yet accurately timed information is well suited as a computational input for object tracking. In this letter, visual data processing is performed for each incoming event at the time it arrives. The method provides a continuous and iterative estimation of the geometric transformation between the model and the events representing the tracked object. It can handle isometry, similarities, and affine distortions and allows for unprecedented real-time performance at equivalent frame rates in the kilohertz range on a standard PC. Furthermore, by using the dimension of time that is currently underexploited by most artificial vision systems, the method we present is able to solve ambiguous cases of object occlusions that classical frame-based techniques handle poorly.

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32. The Benefits of Modeling Slack Variables in SVMs

Fengzhen Tang, Peter Ti■o, Pedro Antonio Gutiérrez, Huanhuan Chen
https://doi.org/10.1162/neco_a_00714 | pp. 954-981

In this letter, we explore the idea of modeling slack variables in support vector machine (SVM) approaches. The study is motivated by SVM+, which models the slacks through a smooth correcting function that is determined by additional (privileged) information about the training examples not available in the test phase. We take a closer look at the meaning and consequences of smooth modeling of slacks, as opposed to determining them in an unconstrained manner through the SVM optimization program. To better understand this difference we only allow the determination and modeling of slack values on the same information--that is, using the same training input in the original input space. We also explore whether it is possible to improve classification performance by combining (in a convex combination) the original SVM slacks with the modeled ones. We show experimentally that this approach not only leads to improved generalization performance but also yields more compact, lower-complexity models. Finally, we extend this idea to the context of ordinal regression, where a natural order among the classes exists. The experimental results confirm principal

findings from the binary case.

33. A Study on the Optimal Double Parameters for Steepest Descent with Momentum

Naimin Zhang

https://doi.org/10.1162/neco_a_00710 | pp. 982-1004

This letter presents the stability analysis for two steepest descent algorithms with momentum for quadratic functions. The corresponding local optimal parameters in Torii and Hagan (2002) and Zhang (2013) are extended to the global optimal parameters, that is, both the optimal learning rates and the optimal momentum factors are obtained simultaneously which make for the fastest convergence.

34. Self-Organization of Control Circuits for Invariant Fiber Projections

Tomas Fernandes, Christoph von der Malsburg

https://doi.org/10.1162/neco_a_00725 | pp. 1005-1032

Assuming that patterns in memory are represented as two-dimensional arrays of local features, just as they are in primary visual cortices, pattern recognition can take the form of elastic graph matching (Lades et al., 1993). Neural implementation of this may be based on preorganized fiber projections that can be activated rapidly with the help of control units (Wolfrum, Wolff, Lücke, & von der Malsburg, 2008). Each control unit governs a set of projection fibers that form part of a coherent mapping. We describe a mathematical model for the ontogenesis of the underlying connectivity based on a principle of network self-organization as described by the Häussler system (Häussler & von der Malsburg, 1983), modified to be sensitive to pattern similarity and to support formation of multiple mappings, each under the command of a control unit. The process takes the form of a soft-winner-take-all, where units compete for the representation of maps. We show simulations for invariant point-to-point and feature-to-feature mappings.

35. A Computationally Efficient Method for Incorporating Spike Waveform Information into Decoding Algorithms

Valérie Ventura, Sonia Todorova

https://doi.org/10.1162/neco_a_00731 | pp. 1033-1050

Spike-based brain-computer interfaces (BCIs) have the potential to restore motor ability to people with paralysis and amputation, and have shown impressive performance in the lab. To transition BCI devices from the lab to the clinic, decoding must proceed automatically and in real time, which prohibits the use of algorithms that are computationally intensive or require manual tweaking. A common choice is to avoid spike sorting and treat the signal on each electrode as if it came from a single neuron, which is fast, easy, and therefore desirable for clinical use. But this approach ignores the kinematic information provided by individual neurons recorded on the same electrode. The contribution of this letter is a linear decoding model that extracts kinematic information from individual neurons without spike-sorting the electrode signals. The method relies on modeling sample averages of waveform features as functions of kinematics, which is automatic and requires minimal data storage and computation. In offline reconstruction of arm trajectories of

a nonhuman primate performing reaching tasks, the proposed method performs as well as decoders based on expertly manually and automatically sorted spikes.

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36. Coding Accuracy Is Not Fully Determined by the Neuronal Model

Lubomir Kostal, Petr Lánský

https://doi.org/10.1162/neco_a_00724 | pp. 1051-1057

It is automatically assumed that the accuracy with which a stimulus can be decoded is entirely determined by the properties of the neuronal system. We challenge this perspective by showing that the identification of pure tone intensities in an auditory nerve fiber depends on both the stochastic response model and the arbitrarily chosen stimulus units. We expose an apparently paradoxical situation in which it is impossible to decide whether loud or quiet tones are encoded more precisely. Our conclusion reaches beyond the topic of auditory neuroscience, however, as we show that the choice of stimulus scale is an integral part of the neural coding problem and not just a matter of convenience.

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37. Solving Stereo Transparency with an Extended Coarse-to-Fine Disparity Energy Model

Zhe Li, Ning Qian

https://doi.org/10.1162/neco_a_00722 | pp. 1058-1082

Modeling stereo transparency with physiologically plausible mechanisms is challenging because in such frameworks, large receptive fields mix up overlapping disparities, whereas small receptive fields can reliably compute only small disparities. It seems necessary to combine information across scales. A coarse-to-fine disparity energy model, with both position- and phase-shift receptive fields, has already been proposed. However, because each scale decodes only one disparity for each location and uses the decoded disparity to select cells at the next scale, this model cannot represent overlapping surfaces at different depths. We have extended the model to solve stereo transparency. First, we introduce multiplicative connections from cells at one scale to the next to implement coarse-to-fine computation. The connection is the strongest when the presynaptic cell's preferred disparity matches the postsynaptic cell's position-shift parameter, encouraging the next scale to encode residual disparities with the more reliable phase-shift mechanism. This modification not only eliminates the artificial decoding and selection steps of the original model but also enables maintenance of complete population responses throughout the coarse-to-fine process. Second, because of this modification, explicit decoding is no longer necessary but rather is for visualization only. We use a simple threshold criterion to decode multiple disparities from population energy responses instead of a single disparity in the original model. We demonstrate our model using simulations on a variety of transparent and nontransparent stereograms. The model also reproduces psychophysically observed disparity interactions (averaging, thickening, attraction, and repulsion) as the depth separation between two overlapping planes varies.

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38. Simultaneous Multichannel Signal Transfers via Chaos in a Recurrent Neural Network

Ken-ichiro Soma, Ryota Mori, Ryuichi Sato, Noriyuki Furumai, Shigetoshi Nara
https://doi.org/10.1162/neco_a_00715 | pp. 1083-1101

We propose neural network model that demonstrates the phenomenon of signal transfer between separated neuron groups via other chaotic neurons that show no apparent correlations with the input signal. The model is a recurrent neural network in which it is supposed that synchronous behavior between small groups of input and output neurons has been learned as fragments of high-dimensional memory patterns, and depletion of neural connections results in chaotic wandering dynamics. Computer experiments show that when a strong oscillatory signal is applied to an input group in the chaotic regime, the signal is successfully transferred to the corresponding output group, although no correlation is observed between the input signal and the intermediary neurons. Signal transfer is also observed when multiple signals are applied simultaneously to separate input groups belonging to different memory attractors. In this sense simultaneous multichannel communications are realized, and the chaotic neural dynamics acts as a signal transfer medium in which the signal appears to be hidden.

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39. Input-Anticipating Critical Reservoirs Show Power Law Forgetting of Unexpected Input Events

Norbert Michael Mayer
https://doi.org/10.1162/neco_a_00730 | pp. 1102-1119

Usually reservoir computing shows an exponential memory decay. This letter investigates under which circumstances echo state networks can show a power law forgetting. That means traces of earlier events can be found in the reservoir for very long time spans. Such a setting requires critical connectivity exactly at the limit of what is permissible according to the echo state condition. However, for general matrices, the limit cannot be determined exactly from theory. In addition, the behavior of the network is strongly influenced by the input flow. Results are presented that use certain types of restricted recurrent connectivity and anticipation learning with regard to the input, where power law forgetting can indeed be achieved.

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40. Learning Dynamics of a Single Polar Variable Complex-Valued Neuron

Tohru Nitta
https://doi.org/10.1162/neco_a_00729 | pp. 1120-1141

This letter investigates the characteristics of the complex-valued neuron model with parameters represented by polar coordinates (called polar variable complex-valued neuron). The parameters of the polar variable complex-valued neuron are unidentifiable. The plateau phenomenon can occur during learning of the polar variable complex-valued neuron. Furthermore, computer simulations suggest that a single polar variable complex-valued neuron has the following characteristics in the case of using the steepest gradient-descent method with square error: (1) unidentifiable parameters (singular points) degrade the learning speed and (2) a plateau can occur during learning. When the weight is attracted to the singular point, the learning tends to become stuck. However, computer simulations also show that the steepest gradient-descent method with amplitude-phase error and the complex-valued natural gradient method could reduce the effects of the singular points. The learning dynamics near singular points depends on the error functions and the training algorithms used.

41. Online EM with Weight-Based Forgetting

Enric Celaya, Alejandro Agostini

https://doi.org/10.1162/neco_a_00723 | pp. 1142-1157

In the online version of the EM algorithm introduced by Sato and Ishii (2000), a time-dependent discount factor is introduced for forgetting the effect of the old estimated values obtained with an earlier, inaccurate estimator. In their approach, forgetting is uniformly applied to the estimators of each mixture component depending exclusively on time, irrespective of the weight attributed to each unit for the observed sample. This causes an excessive forgetting in the less frequently sampled regions. To address this problem, we propose a modification of the algorithm that involves a weight-dependent forgetting, different for each mixture component, in which old observations are forgotten according to the actual weight of the new samples used to replace older values. A comparison of the time-dependent versus the weight-dependent approach shows that the latter improves the accuracy of the approximation and exhibits much greater stability.

42. Homeostatic Plasticity for Single Node Delay-Coupled Reservoir Computing

Hazem Toutounji, Johannes Schumacher, Gordon Pipa

https://doi.org/10.1162/neco_a_00737 | pp. 1159-1185

Supplementing a differential equation with delays results in an infinite-dimensional dynamical system. This property provides the basis for a reservoir computing architecture, where the recurrent neural network is replaced by a single nonlinear node, delay-coupled to itself. Instead of the spatial topology of a network, subunits in the delay-coupled reservoir are multiplexed in time along one delay span of the system. The computational power of the reservoir is contingent on this temporal multiplexing. Here, we learn optimal temporal multiplexing by means of a biologically inspired homeostatic plasticity mechanism. Plasticity acts locally and changes the distances between the subunits along the delay, depending on how responsive these subunits are to the input. After analytically deriving the learning mechanism, we illustrate its role in improving the reservoir's computational power. To this end, we investigate, first, the increase of the reservoir's memory capacity. Second, we predict a NARMA-10 time series, showing that plasticity reduces the normalized root-mean-square error by more than 20%. Third, we discuss plasticity's influence on the reservoir's input-information capacity, the coupling strength between subunits, and the distribution of the readout coefficients.

43. Surrogate Population Models for Large-Scale Neural Simulations

Bryan Tripp

https://doi.org/10.1162/neco_a_00734 | pp. 1186-1222

Because different parts of the brain have rich interconnections, it is not possible to model small parts realistically in isolation. However, it is also impractical to simulate large neural systems in detail. This article outlines a new approach to multiscale modeling of neural systems that involves constructing efficient surrogate models of populations. Given a population of neuron models with correlated activity and with

specific, nonrandom connections, a surrogate model is constructed in order to approximate the aggregate outputs of the population. The surrogate model requires less computation than the neural model, but it has a clear and specific relationship with the neural model. For example, approximate spike rasters for specific neurons can be derived from a simulation of the surrogate model. This article deals specifically with neural engineering framework (NEF) circuits of leaky-integrate-and-fire point neurons. Weighted sums of spikes are modeled by interpolating over latent variables in the population activity, and linear filters operate on gaussian random variables to approximate spike-related fluctuations. It is found that the surrogate models can often closely approximate network behavior with orders-of-magnitude reduction in computational demands, although there are certain systematic differences between the spiking and surrogate models. Since individual spikes are not modeled, some simulations can be performed with much longer steps sizes (e.g., 20 ms). Possible extensions to non-NEF networks and to more complex neuron models are discussed.

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44. Regulation of Local Ambient GABA Levels via Transporter-Mediated GABA Import and Export for Subliminal Learning

Osamu Hoshino

https://doi.org/10.1162/neco_a_00733 | pp. 1223-1251

Perception of supraliminal stimuli might in general be reflected in bursts of action potentials (spikes), and their memory traces could be formed through spike-timing-dependent plasticity (STDP). Memory traces for subliminal stimuli might be formed in a different manner, because subliminal stimulation evokes a fraction (but not a burst) of spikes. Simulations of a cortical neural network model showed that a subliminal stimulus that was too brief (10 msec) to perceive transiently (more than about 500 msec) depolarized stimulus-relevant principal cells and hyperpolarized stimulus-irrelevant principal cells in a subthreshold manner. This led to a small increase or decrease in ongoing-spontaneous spiking activity frequency (less than 1 Hz). Synaptic modification based on STDP during this period effectively enhanced relevant synaptic weights, by which subliminal learning was improved. GABA transporters on GABAergic interneurons modulated local levels of ambient GABA. Ambient GABA molecules acted on extrasynaptic receptors, provided principal cells with tonic inhibitory currents, and contributed to achieving the subthreshold neuronal state. We suggest that ongoing-spontaneous synaptic alteration through STDP following subliminal stimulation may be a possible neuronal mechanism for leaving its memory trace in cortical circuitry. Regulation of local ambient GABA levels by transporter-mediated GABA import and export may be crucial for subliminal learning.

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45. Cortical Spatiotemporal Dimensionality Reduction for Visual Grouping

Giacomo Cocci, Davide Barbieri, Giovanna Citti, Alessandro Sarti

https://doi.org/10.1162/neco_a_00738 | pp. 1252-1293

The visual systems of many mammals, including humans, are able to integrate the geometric information of visual stimuli and perform cognitive tasks at the first stages of the cortical processing. This is thought to be the result of a combination of mechanisms, which include feature extraction at the single cell level and geometric processing by means of cell connectivity. We present a geometric model of such connectivities in the space of detected features associated with spatiotemporal visual stimuli and show how they can be used to obtain low-level object segmentation. The main idea is to define a spectral clustering procedure with

anisotropic affinities over data sets consisting of embeddings of the visual stimuli into higher-dimensional spaces. Neural plausibility of the proposed arguments will be discussed.

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46. Refined Generalization Bounds of Gradient Learning over Reproducing Kernel Hilbert Spaces

Shaogao Lv

https://doi.org/10.1162/neco_a_00739 | pp. 1294-1320

Gradient learning (GL), initially proposed by Mukherjee and Zhou (2006) has been proved to be a powerful tool for conducting variable selection and dimensional reduction simultaneously. This approach presents a nonparametric version of a gradient estimator with positive definite kernels without estimating the true function itself, so that the proposed version has wide applicability and allows for complex effects between predictors. In terms of theory, however, existing generalization bounds for GL depend on capacity-independent techniques, and the capacity of kernel classes cannot be characterized completely. Thus, this letter considers GL estimators that minimize the empirical convex risk. We prove generalization bounds for such estimators with rates that are faster than previous results. Moreover, we provide a novel upper bound for Rademacher chaos complexity of order two, which also plays an important role in general pairwise-type estimations, including ranking and score problems.

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47. Timescale Separation in Recurrent Neural Networks

Thomas R. Flynn

https://doi.org/10.1162/neco_a_00740 | pp. 1321-1344

Supervised learning in recurrent neural networks involves two processes: the neuron activity from which gradients are estimated and the process on connection parameters induced by these measurements. A problem such algorithms must address is how to balance the relative rates of these activities so that accurate sensitivity estimates are obtained while still allowing synaptic modification to take place at a rate sufficient for learning. We show how to calculate a sufficient timescale separation between these two processes for a class of contracting neural networks.

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48. FastMMD: Ensemble of Circular Discrepancy for Efficient Two-Sample Test

Ji Zhao, Deyu Meng

https://doi.org/10.1162/neco_a_00732 | pp. 1345-1372

The maximum mean discrepancy (MMD) is a recently proposed test statistic for the two-sample test. Its quadratic time complexity, however, greatly hampers its availability to large-scale applications. To accelerate the MMD calculation, in this study we propose an efficient method called FastMMD. The core idea of FastMMD is to equivalently transform the MMD with shift-invariant kernels into the amplitude expectation of a linear combination of sinusoid components based on Bochner's theorem and Fourier transform (Rahimi & Recht, 2007). Taking advantage of sampling the Fourier transform, FastMMD decreases the time complexity for MMD calculation from O(N(2)d) to O(LN d), where N and d are the size

and dimension of the sample set, respectively. Here, L is the number of basis functions for approximating kernels that determines the approximation accuracy. For kernels that are spherically invariant, the computation can be further accelerated to O(LN log d) by using the Fastfood technique (Le, Sarlós, & Smola, 2013). The uniform convergence of our method has also been theoretically proved in both unbiased and biased estimates. We also provide a geometric explanation for our method, ensemble of circular discrepancy, which helps us understand the insight of MMD and we hope will lead to more extensive metrics for assessing the two-sample test task. Experimental results substantiate that the accuracy of FastMMD is similar to that of MMD and with faster computation and lower variance than existing MMD approximation methods.

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49. Unifying Blind Separation and Clustering for Resting-State EEG/MEG Functional Connectivity Analysis

Jun-ichiro Hirayama, Takeshi OGAWA, Aapo Hyvärinen

https://doi.org/10.1162/neco_a_00747 | pp. 1373-1404

Unsupervised analysis of the dynamics (nonstationarity) of functional brain connectivity during rest has recently received a lot of attention in the neuroimaging and neuroengineering communities. Most studies have used functional magnetic resonance imaging, but electroencephalography (EEG) and magnetoencephalography (MEG) also hold great promise for analyzing nonstationary functional connectivity with high temporal resolution. Previous EEG/MEG analyses divided the problem into two consecutive stages: the separation of neural sources and then the connectivity analysis of the separated sources. Such nonoptimal division into two stages may bias the result because of the different prior assumptions made about the data in the two stages. We propose a unified method for separating EEG/MEG sources and learning their functional connectivity (coactivation) patterns. We combine blind source separation (BSS) with unsupervised clustering of the activity levels of the sources in a single probabilistic model. A BSS is performed on the Hilbert transforms of band-limited EEG/MEG signals, and coactivation patterns are learned by a mixture model of source envelopes. Simulation studies show that the unified approach often outperforms conventional two-stage methods, indicating further the benefit of using Hilbert transforms to deal with oscillatory sources. Experiments on resting-state EEG data, acquired in conjunction with a cued motor imagery or nonimagery task, also show that the states (clusters) obtained by the proposed method often correlate better with physiologically meaningful quantities than those obtained by a two-stage method.

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50. Attentional Bias Through Oscillatory Coherence Between Excitatory Activity and Inhibitory Minima

Sebastian Blaes, Thomas Burwick

https://doi.org/10.1162/neco_a_00742 | pp. 1405-1437

An implementation of attentional bias is presented for a network model that couples excitatory and inhibitory oscillatory units in a manner that is inspired by the mechanisms that generate cortical gamma oscillations. Attentional biases are implemented as oscillatory coherences between excitatory units that encode the spatial location or features of the target and the pool of inhibitory units. This form of attentional bias is motivated by neurophysiological findings that relate selective attention to spike field coherence. Including also pattern recognition mechanisms, we demonstrate how this implementation of attentional bias leads to

selection of an attentional target while suppressing distracters for cases of spatial and feature-based attention. With respect to neurophysiological observations, we argue that the recently found positive correlation between high firing rates and strong gamma locking with attention (Vinck, Womelsdorf, Buffalo, Desimone, & Fries, 2013) may point to an essential mechanism of the brain's attentional selection and suppression processes.

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51. Clusterless Decoding of Position from Multiunit Activity Using a Marked Point Process Filter

Xinyi Deng, Daniel F. Liu, Kenneth Kay, Loren M. Frank, Uri T. Eden

https://doi.org/10.1162/neco_a_00744 | pp. 1438-1460

Point process filters have been applied successfully to decode neural signals and track neural dynamics. Traditionally these methods assume that multiunit spiking activity has already been correctly spike-sorted. As a result, these methods are not appropriate for situations where sorting cannot be performed with high precision, such as real-time decoding for brain-computer interfaces. Because the unsupervised spike-sorting problem remains unsolved, we took an alternative approach that takes advantage of recent insights into clusterless decoding. Here we present a new point process decoding algorithm that does not require multiunit signals to be sorted into individual units. We use the theory of marked point processes to construct a function that characterizes the relationship between a covariate of interest (in this case, the location of a rat on a track) and features of the spike waveforms. In our example, we use tetrode recordings, and the marks represent a four-dimensional vector of the maximum amplitudes of the spike waveform on each of the four electrodes. In general, the marks may represent any features of the spike waveform. We then use Bayes's rule to estimate spatial location from hippocampal neural activity. We validate our approach with a simulation study and experimental data recorded in the hippocampus of a rat moving through a linear environment. Our decoding algorithm accurately reconstructs the rat's position from unsorted multiunit spiking activity. We then compare the quality of our decoding algorithm to that of a traditional spike-sorting and decoding algorithm. Our analyses show that the proposed decoding algorithm performs equivalent to or better than algorithms based on sorted single-unit activity. These results provide a path toward accurate real-time decoding of spiking patterns that could be used to carry out content-specific manipulations of population activity in hippocampus or elsewhere in the brain.

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52. A Hebbian/Anti-Hebbian Neural Network for Linear Subspace Learning: A Derivation from Multidimensional Scaling of Streaming Data

Cengiz Pehlevan, Tao Hu, Dmitri B. Chklovskii

https://doi.org/10.1162/neco_a_00745 | pp. 1461-1495

Neural network models of early sensory processing typically reduce the dimensionality of streaming input data. Such networks learn the principal subspace, in the sense of principal component analysis, by adjusting synaptic weights according to activity-dependent learning rules. When derived from a principled cost function, these rules are nonlocal and hence biologically implausible. At the same time, biologically plausible local rules have been postulated rather than derived from a principled cost function. Here, to bridge this gap, we derive a biologically plausible network for subspace learning on streaming data by minimizing a principled cost function. In a departure from previous work, where cost was quantified by the representation,

or reconstruction, error, we adopt a multidimensional scaling cost function for streaming data. The resulting algorithm relies only on biologically plausible Hebbian and anti-Hebbian local learning rules. In a stochastic setting, synaptic weights converge to a stationary state, which projects the input data onto the principal subspace. If the data are generated by a nonstationary distribution, the network can track the principal subspace. Thus, our result makes a step toward an algorithmic theory of neural computation.

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53. Learning Slowness in a Sparse Model of Invariant Feature Detection

Thusitha N. Chandrapala, Bertram E. Shi

https://doi.org/10.1162/neco_a_00743 | pp. 1496-1529

Primary visual cortical complex cells are thought to serve as invariant feature detectors and to provide input to higher cortical areas. We propose a single model for learning the connectivity required by complex cells that integrates two factors that have been hypothesized to play a role in the development of invariant feature detectors: temporal slowness and sparsity. This model, the generative adaptive subspace self-organizing map (GASSOM), extends Kohonen's adaptive subspace self-organizing map (ASSOM) with a generative model of the input. Each observation is assumed to be generated by one among many nodes in the network, each being associated with a different subspace in the space of all observations. The generating nodes evolve according to a first-order Markov chain and generate inputs that lie close to the associated subspace. This model differs from prior approaches in that temporal slowness is not an externally imposed criterion to be maximized during learning but, rather, an emergent property of the model structure as it seeks a good model of the input statistics. Unlike the ASSOM, the GASSOM does not require an explicit segmentation of the input training vectors into separate episodes. This enables us to apply this model to an unlabeled naturalistic image sequence generated by a realistic eye movement model. We show that the emergence of temporal slowness within the model improves the invariance of feature detectors trained on this input.

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54. On the Spike Train Variability Characterized by Variance-to-Mean Power Relationship

Shinsuke Koyama

https://doi.org/10.1162/neco_a_00748 | pp. 1530-1548

We propose a statistical method for modeling the non-Poisson variability of spike trains observed in a wide range of brain regions. Central to our approach is the assumption that the variance and the mean of interspike intervals are related by a power function characterized by two parameters: the scale factor and exponent. It is shown that this single assumption allows the variability of spike trains to have an arbitrary scale and various dependencies on the firing rate in the spike count statistics, as well as in the interval statistics, depending on the two parameters of the power function. We also propose a statistical model for spike trains that exhibits the variance-to-mean power relationship. Based on this, a maximum likelihood method is developed for inferring the parameters from rate-modulated spike trains. The proposed method is illustrated on simulated and experimental spike trains.

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55. Comments and Correction on ”U-Processes and Preference Learning” (Neural Computation Vol. 26, pp. 2896–2924, 2014)

Wojciech Rejchel, Hong Li, Chuanbao Ren, Luoqing Li

https://doi.org/10.1162/neco_c_00746 | pp. 1549-1553

This note corrects an error in the proof of corollary 1 of Li et al. (2014). The original claim of the contraction principle in appendix D of Li et al. no longer holds.

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56. A Statistical Framework to Infer Delay and Direction of Information Flow from Measurements of Complex Systems

Johannes Schumacher, Thomas Wunderle, Pascal Fries, Frank Jäkel, Gordon Pipa

https://doi.org/10.1162/neco_a_00756 | pp. 1555-1608

In neuroscience, data are typically generated from neural network activity. The resulting time series represent measurements from spatially distributed subsystems with complex interactions, weakly coupled to a high-dimensional global system. We present a statistical framework to estimate the direction of information flow and its delay in measurements from systems of this type. Informed by differential topology, gaussian process regression is employed to reconstruct measurements of putative driving systems from measurements of the driven systems. These reconstructions serve to estimate the delay of the interaction by means of an analytical criterion developed for this purpose. The model accounts for a range of possible sources of uncertainty, including temporally evolving intrinsic noise, while assuming complex nonlinear dependencies. Furthermore, we show that if information flow is delayed, this approach also allows for inference in strong coupling scenarios of systems exhibiting synchronization phenomena. The validity of the method is demonstrated with a variety of delay-coupled chaotic oscillators. In addition, we show that these results seamlessly transfer to local field potentials in cat visual cortex.

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57. An Empirical Model for Reliable Spiking Activity

Wanjie Wang, Shreejoy J. Tripathy, Krishnan Padmanabhan, Nathaniel N. Urban, Robert E. Kass

https://doi.org/10.1162/neco_a_00754 | pp. 1609-1623

Understanding a neuron's transfer function, which relates a neuron's inputs to its outputs, is essential for understanding the computational role of single neurons. Recently, statistical models, based on point processes and using generalized linear model (GLM) technology, have been widely applied to predict dynamic neuronal transfer functions. However, the standard version of these models fails to capture important features of neural activity, such as responses to stimuli that elicit highly reliable trial-to-trial spiking. Here, we consider a generalization of the usual GLM that incorporates nonlinearity by modeling reliable and nonreliable spikes as being generated by distinct stimulus features. We develop and apply these models to spike trains from olfactory bulb mitral cells recorded in vitro. We find that spike generation in these neurons is better modeled when reliable and unreliable spikes are considered separately and that this effect is most pronounced for neurons with a large number of both reliable and unreliable spikes.

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58. Inheritance of Hippocampal Place Fields Through Hebbian Learning: Effects of Theta Modulation and Phase Precession on Structure Formation

Tiziano D'Albis, Jorge Jaramillo, Henning Sprekeler, Richard Kempter

https://doi.org/10.1162/neco_a_00752 | pp. 1624-1672

A place cell is a neuron that fires whenever the animal traverses a particular location of the environment-the place field of the cell. Place cells are found in two regions of the rodent hippocampus: CA3 and CA1. Motivated by the anatomical connectivity between these two regions and by the evidence for synaptic plasticity at these connections, we study how a place field in CA1 can be inherited from an upstream region such as CA3 through a Hebbian learning rule, in particular, through spike-timing-dependent plasticity (STDP). To this end, we model a population of CA3 place cells projecting to a single CA1 cell, and we assume that the CA1 input synapses are plastic according to STDP. With both numerical and analytical methods, we show that in the case of overlapping CA3 input place fields, the STDP learning rule leads to the formation of a place field in CA1. We then investigate the roles of the hippocampal theta modulation and phase precession on the inheritance process. We find that theta modulation favors the inheritance and leads to faster place field formation whereas phase precession changes the drift of CA1 place fields over time.

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59. Competitive STDP Learning of Overlapping Spatial Patterns

Dalius Krunglevicius

https://doi.org/10.1162/neco_a_00753 | pp. 1673-1685

Spike-timing-dependent plasticity (STDP) is a set of Hebbian learning rules firmly based on biological evidence. It has been demonstrated that one of the STDP learning rules is suited for learning spatiotemporal patterns. When multiple neurons are organized in a simple competitive spiking neural network, this network is capable of learning multiple distinct patterns. If patterns overlap significantly (i.e., patterns are mutually inclusive), however, competition would not preclude trained neuron's responding to a new pattern and adjusting synaptic weights accordingly. This letter presents a simple neural network that combines vertical inhibition and Euclidean distance-dependent synaptic strength factor. This approach helps to solve the problem of pattern size-dependent parameter optimality and significantly reduces the probability of a neuron's forgetting an already learned pattern. For demonstration purposes, the network was trained for the first ten letters of the Braille alphabet.

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60. A Reward-Maximizing Spiking Neuron as a Bounded Rational Decision Maker

Felix Leibfried, Daniel A. Braun

https://doi.org/10.1162/neco_a_00758 | pp. 1686-1720

Rate distortion theory describes how to communicate relevant information most efficiently over a channel with limited capacity. One of the many applications of rate distortion theory is bounded rational decision making, where decision makers are modeled as information channels that transform sensory input into motor output under the constraint that their channel capacity is limited. Such a bounded rational decision maker can be thought to optimize an objective function that trades off the decision maker's utility or cumulative reward against the information processing cost measured by the mutual information between

sensory input and motor output. In this study, we interpret a spiking neuron as a bounded rational decision maker that aims to maximize its expected reward under the computational constraint that the mutual information between the neuron's input and output is upper bounded. This abstract computational constraint translates into a penalization of the deviation between the neuron's instantaneous and average firing behavior. We derive a synaptic weight update rule for such a rate distortion optimizing neuron and show in simulations that the neuron efficiently extracts reward-relevant information from the input by trading off its synaptic strengths against the collected reward.

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61. Optimality of Upper-Arm Reaching Trajectories Based on the Expected Value of the Metabolic Energy Cost

Yoshiaki Taniai, Jun Nishii

https://doi.org/10.1162/neco_a_00757 | pp. 1721-1737

When we move our body to perform a movement task, our central nervous system selects a movement trajectory from an infinite number of possible trajectories under constraints that have been acquired through evolution and learning. Minimization of the energy cost has been suggested as a potential candidate for a constraint determining locomotor parameters, such as stride frequency and stride length; however, other constraints have been proposed for a human upper-arm reaching task. In this study, we examined whether the minimum metabolic energy cost model can also explain the characteristics of the upper-arm reaching trajectories. Our results show that the optimal trajectory that minimizes the expected value of energy cost under the effect of signal-dependent noise on motor commands expresses not only the characteristics of reaching movements of typical speed but also those of slower movements. These results suggest that minimization of the energy cost would be a basic constraint not only in locomotion but also in upper-arm reaching.

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62. Active Learning Using Hint Information

Chunliang Li, Chun-Sung Ferng, Hsuan-Tien Lin

https://doi.org/10.1162/neco_a_00755 | pp. 1738-1765

The abundance of real-world data and limited labeling budget calls for active learning, an important learning paradigm for reducing human labeling efforts. Many recently developed active learning algorithms consider both uncertainty and representativeness when making querying decisions. However, exploiting representativeness with uncertainty concurrently usually requires tackling sophisticated and challenging learning tasks, such as clustering. In this letter, we propose a new active learning framework, called hinted sampling, which takes both uncertainty and representativeness into account in a simpler way. We design a novel active learning algorithm within the hinted sampling framework with an extended support vector machine. Experimental results validate that the novel active learning algorithm can result in a better and more stable performance than that achieved by state-of-the-art algorithms. We also show that the hinted sampling framework allows improving another active learning algorithm designed from the transductive support vector machine.

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63. Subsampled Hessian Newton Methods for Supervised Learning

Chien-Chih Wang, Huang Chun-Heng, Chih■Jen Lin

https://doi.org/10.1162/neco_a_00751 | pp. 1766-1795

Newton methods can be applied in many supervised learning approaches. However, for large-scale data, the use of the whole Hessian matrix can be time-consuming. Recently, subsampled Newton methods have been proposed to reduce the computational time by using only a subset of data for calculating an approximation of the Hessian matrix. Unfortunately, we find that in some situations, the running speed is worse than the standard Newton method because cheaper but less accurate search directions are used. In this work, we propose some novel techniques to improve the existing subsampled Hessian Newton method. The main idea is to solve a two-dimensional subproblem per iteration to adjust the search direction to better minimize the second-order approximation of the function value. We prove the theoretical convergence of the proposed method. Experiments on logistic regression, linear SVM, maximum entropy, and deep networks indicate that our techniques significantly reduce the running time of the subsampled Hessian Newton method. The resulting algorithm becomes a compelling alternative to the standard Newton method for large-scale data classification.

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64. Minimal Sign Representation of Boolean Functions: Algorithms and Exact Results for Low Dimensions

Can Eren Sezener, Erhan Öztop

https://doi.org/10.1162/neco_a_00750 | pp. 1796-1823

Boolean functions (BFs) are central in many fields of engineering and mathematics, such as cryptography, circuit design, and combinatorics. Moreover, they provide a simple framework for studying neural computation mechanisms of the brain. Many representation schemes for BFs exist to satisfy the needs of the domain they are used in. In neural computation, it is of interest to know how many input lines a neuron would need to represent a given BF. A common BF representation to study this is the so-called polynomial sign representation where [Formula: see text] and 1 are associated with true and false, respectively. The polynomial is treated as a real-valued function and evaluated at its parameters, and the sign of the polynomial is then taken as the function value. The number of input lines for the modeled neuron is exactly the number of terms in the polynomial. This letter investigates the minimum number of terms, that is, the minimum threshold density, that is sufficient to represent a given BF and more generally aims to find the maximum over this quantity for all BFs in a given dimension. With this work, for the first time exact results for four- and five-variable BFs are obtained, and strong bounds for six-variable BFs are derived. In addition, some connections between the sign representation framework and bent functions are derived, which are generally studied for their desirable cryptographic properties.

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65. Extracting Low-Dimensional Latent Structure from Time Series in the Presence of Delays

Karthik C. Lakshmanan, Patrick T. Sadtler, Elizabeth C. Tyler■Kabara, Aaron P. Batista, Byron M. Yu

https://doi.org/10.1162/neco_a_00759 | pp. 1825-1856

Noisy, high-dimensional time series observations can often be described by a set of low-dimensional latent variables. Commonly used methods to extract these latent variables typically assume instantaneous relationships between the latent and observed variables. In many physical systems, changes in the latent variables manifest as changes in the observed variables after time delays. Techniques that do not account for these delays can recover a larger number of latent variables than are present in the system, thereby making the latent representation more difficult to interpret. In this work, we introduce a novel probabilistic technique, time-delay gaussian-process factor analysis (TD-GPFA), that performs dimensionality reduction in the presence of a different time delay between each pair of latent and observed variables. We demonstrate how using a gaussian process to model the evolution of each latent variable allows us to tractably learn these delays over a continuous domain. Additionally, we show how TD-GPFA combines temporal smoothing and dimensionality reduction into a common probabilistic framework. We present an expectation/conditional maximization either (ECME) algorithm to learn the model parameters. Our simulations demonstrate that when time delays are present, TD-GPFA is able to correctly identify these delays and recover the latent space. We then applied TD-GPFA to the activity of tens of neurons recorded simultaneously in the macaque motor cortex during a reaching task. TD-GPFA is able to better describe the neural activity using a more parsimonious latent space than GPFA, a method that has been used to interpret motor cortex data but does not account for time delays. More broadly, TD-GPFA can help to unravel the mechanisms underlying high-dimensional time series data by taking into account physical delays in the system.

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66. Is First-Order Vector Autoregressive Model Optimal for fMRI Data?

Chee■Ming Ting, Abd■Krim Seghouane, Muhammad Usman Khalid, Sh■Hussain Salleh

https://doi.org/10.1162/neco_a_00765 | pp. 1857-1871

We consider the problem of selecting the optimal orders of vector autoregressive (VAR) models for fMRI data. Many previous studies used model order of one and ignored that it may vary considerably across data sets depending on different data dimensions, subjects, tasks, and experimental designs. In addition, the classical information criteria (IC) used (e.g., the Akaike IC (AIC)) are biased and inappropriate for the high-dimensional fMRI data typically with a small sample size. We examine the mixed results on the optimal VAR orders for fMRI, especially the validity of the order-one hypothesis, by a comprehensive evaluation using different model selection criteria over three typical data types--a resting state, an event-related design, and a block design data set--with varying time series dimensions obtained from distinct functional brain networks. We use a more balanced criterion, Kullback's IC (KIC) based on Kullback's symmetric divergence combining two directed divergences. We also consider the bias-corrected versions (AICc and KICc) to improve VAR model selection in small samples. Simulation results show better small-sample selection performance of the proposed criteria over the classical ones. Both bias-corrected ICs provide more accurate and consistent model order choices than their biased counterparts, which suffer from overfitting, with KICc performing the best. Results on real data show that orders greater than one were selected by all criteria across all data sets for the small to moderate dimensions, particularly from small, specific networks such as the resting-state default mode network and the task-related motor networks, whereas low orders close to one but not necessarily one were chosen for the large dimensions of full-brain networks.

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67. A Novel Reconstruction Framework for Time-Encoded Signals with Integrate-and-Fire Neurons

Dorian Florescu, Daniel Coca

https://doi.org/10.1162/neco_a_00764 | pp. 1872-1898

Integrate-and-fire neurons are time encoding machines that convert the amplitude of an analog signal into a nonuniform, strictly increasing sequence of spike times. Under certain conditions, the encoded signals can be reconstructed from the nonuniform spike time sequences using a time decoding machine. Time encoding and time decoding methods have been studied using the nonuniform sampling theory for band-limited spaces, as well as for generic shift-invariant spaces. This letter proposes a new framework for studying IF time encoding and decoding by reformulating the IF time encoding problem as a uniform sampling problem. This framework forms the basis for two new algorithms for reconstructing signals from spike time sequences. We demonstrate that the proposed reconstruction algorithms are faster, and thus better suited for real-time processing, while providing a similar level of accuracy, compared to the standard reconstruction algorithm.

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68. Online Direct Density-Ratio Estimation Applied to Inlier-Based Outlier Detection

Marthinus Christoffel du Plessis, Hiroaki Shiino, Masashi Sugiyama

https://doi.org/10.1162/neco_a_00761 | pp. 1899-1914

Many machine learning problems, such as nonstationarity adaptation, outlier detection, dimensionality reduction, and conditional density estimation, can be effectively solved by using the ratio of probability densities. Since the naive two-step procedure of first estimating the probability densities and then taking their ratio performs poorly, methods to directly estimate the density ratio from two sets of samples without density estimation have been extensively studied recently. However, these methods are batch algorithms that use the whole data set to estimate the density ratio, and they are inefficient in the online setup, where training samples are provided sequentially and solutions are updated incrementally without storing previous samples. In this letter, we propose two online density-ratio estimators based on the adaptive regularization of weight vectors. Through experiments on inlier-based outlier detection, we demonstrate the usefulness of the proposed methods.

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69. Relations Among Some Low-Rank Subspace Recovery Models

Hongyang Zhang, Zhouchen Lin, Chao Zhang, Junbin Gao

https://doi.org/10.1162/neco_a_00762 | pp. 1915-1950

Recovering intrinsic low-dimensional subspaces from data distributed on them is a key preprocessing step to many applications. In recent years, a lot of work has modeled subspace recovery as low-rank minimization problems. We find that some representative models, such as robust principal component analysis (R-PCA), robust low-rank representation (R-LRR), and robust latent low-rank representation (R-LatLRR), are actually deeply connected. More specifically, we discover that once a solution to one of the models is obtained, we can obtain the solutions to other models in closed-form formulations. Since R-PCA is the simplest, our discovery makes it the center of low-rank subspace recovery models. Our work has two

important implications. First, R-PCA has a solid theoretical foundation. Under certain conditions, we could find globally optimal solutions to these low-rank models at an overwhelming probability, although these models are nonconvex. Second, we can obtain significantly faster algorithms for these models by solving R-PCA first. The computation cost can be further cut by applying low-complexity randomized algorithms, for example, our novel l2,1 filtering algorithm, to R-PCA. Although for the moment the formal proof of our l2,1 filtering algorithm is not yet available, experiments verify the advantages of our algorithm over other state-of-the-art methods based on the alternating direction method.

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70. A Fast Algorithm for Learning Overcomplete Dictionary for Sparse Representation Based on Proximal Operators

Zhenni Li, Shuxue Ding, Yujie Li

https://doi.org/10.1162/neco_a_00763 | pp. 1951-1982

We present a fast, efficient algorithm for learning an overcomplete dictionary for sparse representation of signals. The whole problem is considered as a minimization of the approximation error function with a coherence penalty for the dictionary atoms and with the sparsity regularization of the coefficient matrix. Because the problem is nonconvex and nonsmooth, this minimization problem cannot be solved efficiently by an ordinary optimization method. We propose a decomposition scheme and an alternating optimization that can turn the problem into a set of minimizations of piecewise quadratic and univariate subproblems, each of which is a single variable vector problem, of either one dictionary atom or one coefficient vector. Although the subproblems are still nonsmooth, remarkably they become much simpler so that we can find a closed-form solution by introducing a proximal operator. This leads to an efficient algorithm for sparse representation. To our knowledge, applying the proximal operator to the problem with an incoherence term and obtaining the optimal dictionary atoms in closed form with a proximal operator technique have not previously been studied. The main advantages of the proposed algorithm are that, as suggested by our analysis and simulation study, it has lower computational complexity and a higher convergence rate than state-of-the-art algorithms. In addition, for real applications, it shows good performance and significant reductions in computational time.

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71. Fusion of Scores in a Detection Context Based on Alpha Integration

Antonio Soriano, Luis Vergara, Ahmed Bouziane, Addisson Salazar

https://doi.org/10.1162/neco_a_00766 | pp. 1983-2010

We present a new method for fusing scores corresponding to different detectors (two-hypotheses case). It is based on alpha integration, which we have adapted to the detection context. Three optimization methods are presented: least mean square error, maximization of the area under the ROC curve, and minimization of the probability of error. Gradient algorithms are proposed for the three methods. Different experiments with simulated and real data are included. Simulated data consider the two-detector case to illustrate the factors influencing alpha integration and demonstrate the improvements obtained by score fusion with respect to individual detector performance. Two real data cases have been considered. In the first, multimodal biometric data have been processed. This case is representative of scenarios in which the probability of detection is to be maximized for a given probability of false alarm. The second case is the automatic analysis of electroencephalogram and electrocardiogram records with the aim of reproducing the medical

expert detections of arousal during sleeping. This case is representative of scenarios in which probability of error is to be minimized. The general superior performance of alpha integration verifies the interest of optimizing the fusing parameters.

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72. Understanding Emergent Dynamics: Using a Collective Activity Coordinate of a Neural Network to Recognize Time-Varying Patterns

J. J. Hopfield

https://doi.org/10.1162/neco_a_00768 | pp. 2011-2038

In higher animals, complex and robust behaviors are produced by the microscopic details of large structured ensembles of neurons. I describe how the emergent computational dynamics of a biologically based neural network generates a robust natural solution to the problem of categorizing time-varying stimulus patterns such as spoken words or animal stereotypical behaviors. The recognition of these patterns is made difficult by their substantial variation in cadence and duration. The neural circuit behaviors used are similar to those associated with brain neural integrators. In the larger context described here, this kind of circuit becomes a building block of an entirely different computational algorithm for solving complex problems. While the network behavior is simulated in detail, a collective view is essential to understanding the results. A closed equation of motion for the collective variable describes an algorithm that quantitatively accounts for many aspects of the emergent network computation. The feedback connections and ongoing activity in the network shape the collective dynamics onto a reduced dimensionality manifold of activity space, which defines the algorithm and computation actually performed. The external inputs are weak and are not the dominant drivers of network activity.

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73. Indefinite Proximity Learning: A Review

Frank-Michael Schleif, Peter Ti■o

https://doi.org/10.1162/neco_a_00770 | pp. 2039-2096

Efficient learning of a data analysis task strongly depends on the data representation. Most methods rely on (symmetric) similarity or dissimilarity representations by means of metric inner products or distances, providing easy access to powerful mathematical formalisms like kernel or branch-and-bound approaches. Similarities and dissimilarities are, however, often naturally obtained by nonmetric proximity measures that cannot easily be handled by classical learning algorithms. Major efforts have been undertaken to provide approaches that can either directly be used for such data or to make standard methods available for these types of data. We provide a comprehensive survey for the field of learning with nonmetric proximities. First, we introduce the formalism used in nonmetric spaces and motivate specific treatments for nonmetric proximity data. Second, we provide a systematization of the various approaches. For each category of approaches, we provide a comparative discussion of the individual algorithms and address complexity issues and generalization properties. In a summarizing section, we provide a larger experimental study for the majority of the algorithms on standard data sets. We also address the problem of large-scale proximity learning, which is often overlooked in this context and of major importance to make the method relevant in practice. The algorithms we discuss are in general applicable for proximity-based clustering, one-class classification, classification, regression, and embedding approaches. In the experimental part, we focus on classification tasks.

74. A Note on Entropy Estimation

Thomas Schürmann

https://doi.org/10.1162/neco_a_00775 | pp. 2097-2106

We compare an entropy estimator H(z) recently discussed by Zhang (2012) with two estimators, H(1) and H(2), introduced by Grassberger (2003) and Schürmann (2004). We prove the identity H(z) ≡ H(1), which has not been taken into account by Zhang (2012). Then we prove that the systematic error (bias) of H(1) is less than or equal to the bias of the ordinary likelihood (or plug-in) estimator of entropy. Finally, by numerical simulation, we verify that for the most interesting regime of small sample estimation and large event spaces, the estimator H(2) has a significantly smaller statistical error than H(z).

75. Recurrent Neural Network Approach Based on the Integral Representation of the Drazin Inverse

Predrag S. Stanimirovi■, Ivan S. Živkovi■, Yimin Wei

https://doi.org/10.1162/neco_a_00771 | pp. 2107-2131

In this letter, we present the dynamical equation and corresponding artificial recurrent neural network for computing the Drazin inverse for arbitrary square real matrix, without any restriction on its eigenvalues. Conditions that ensure the stability of the defined recurrent neural network as well as its convergence toward the Drazin inverse are considered. Several illustrative examples present the results of computer simulations.

76. Visual Categorization with Random Projection

Rosa I. Arriaga, David Rutter, Maya Çakmak, Santosh Vempala

https://doi.org/10.1162/neco_a_00769 | pp. 2132-2147

Humans learn categories of complex objects quickly and from a few examples. Random projection has been suggested as a means to learn and categorize efficiently. We investigate how random projection affects categorization by humans and by very simple neural networks on the same stimuli and categorization tasks, and how this relates to the robustness of categories. We find that (1) drastic reduction in stimulus complexity via random projection does not degrade performance in categorization tasks by either humans or simple neural networks, (2) human accuracy and neural network accuracy are remarkably correlated, even at the level of individual stimuli, and (3) the performance of both is strongly indicated by a natural notion of category robustness.

77. Accuracy and Efficiency in Fixed-Point Neural ODE Solvers

Michael Hopkins, Steve Furber

https://doi.org/10.1162/neco_a_00772 | pp. 2148-2182

Simulation of neural behavior on digital architectures often requires the solution of ordinary differential equations (ODEs) at each step of the simulation. For some neural models, this is a significant computational burden, so efficiency is important. Accuracy is also relevant because solutions can be sensitive to model

parameterization and time step. These issues are emphasized on fixed-point processors like the ARM unit used in the SpiNNaker architecture. Using the Izhikevich neural model as an example, we explore some solution methods, showing how specific techniques can be used to find balanced solutions. We have investigated a number of important and related issues, such as introducing explicit solver reduction (ESR) for merging an explicit ODE solver and autonomous ODE into one algebraic formula, with benefits for both accuracy and speed; a simple, efficient mechanism for cancelling the cumulative lag in state variables caused by threshold crossing between time steps; an exact result for the membrane potential of the Izhikevich model with the other state variable held fixed. Parametric variations of the Izhikevich neuron show both similarities and differences in terms of algorithms and arithmetic types that perform well, making an overall best solution challenging to identify, but we show that particular cases can be improved significantly using the techniques described. Using a 1 ms simulation time step and 32-bit fixed-point arithmetic to promote real-time performance, one of the second-order Runge-Kutta methods looks to be the best compromise; Midpoint for speed or Trapezoid for accuracy. SpiNNaker offers an unusual combination of low energy use and real-time performance, so some compromises on accuracy might be expected. However, with a careful choice of approach, results comparable to those of general-purpose systems should be possible in many realistic cases.

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78. A Simple Label Switching Algorithm for Semisupervised Structural SVMs

P. Balamurugan, Shirish Shevade, S. Sundararajan

https://doi.org/10.1162/neco_a_00773 | pp. 2183-2206

In structured output learning, obtaining labeled data for real-world applications is usually costly, while unlabeled examples are available in abundance. Semisupervised structured classification deals with a small number of labeled examples and a large number of unlabeled structured data. In this work, we consider semisupervised structural support vector machines with domain constraints. The optimization problem, which in general is not convex, contains the loss terms associated with the labeled and unlabeled examples, along with the domain constraints. We propose a simple optimization approach that alternates between solving a supervised learning problem and a constraint matching problem. Solving the constraint matching problem is difficult for structured prediction, and we propose an efficient and effective label switching method to solve it. The alternating optimization is carried out within a deterministic annealing framework, which helps in effective constraint matching and avoiding poor local minima, which are not very useful. The algorithm is simple and easy to implement. Further, it is suitable for any structured output learning problem where exact inference is available. Experiments on benchmark sequence labeling data sets and a natural language parsing data set show that the proposed approach, though simple, achieves comparable generalization performance.

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79. Multiview Bayesian Correlated Component Analysis

Simon Kamronn, Andreas Trier Poulsen, Lars Kai Hansen

https://doi.org/10.1162/neco_a_00774 | pp. 2207-2230

Correlated component analysis as proposed by Dmochowski, Sajda, Dias, and Parra (2012) is a tool for investigating brain process similarity in the responses to multiple views of a given stimulus. Correlated components are identified under the assumption that the involved spatial networks are identical. Here we

propose a hierarchical probabilistic model that can infer the level of universality in such multiview data, from completely unrelated representations, corresponding to canonical correlation analysis, to identical representations as in correlated component analysis. This new model, which we denote Bayesian correlated component analysis, evaluates favorably against three relevant algorithms in simulated data. A well-established benchmark EEG data set is used to further validate the new model and infer the variability of spatial representations across multiple subjects.

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80. Linear Methods for Efficient and Fast Separation of Two Sources Recorded with a Single Microphone

S. Bhargava, Florian Blättler, Sepp Kollmorgen, Shih■Chii Liu, Richard H. R. Hahnloser
https://doi.org/10.1162/neco_a_00776 | pp. 2231-2259

This letter addresses the problem of separating two speakers from a single microphone recording. Three linear methods are tested for source separation, all of which operate directly on sound spectrograms: (1) eigenmode analysis of covariance difference to identify spectro-temporal features associated with large variance for one source and small variance for the other source; (2) maximum likelihood demixing in which the mixture is modeled as the sum of two gaussian signals and maximum likelihood is used to identify the most likely sources; and (3) suppression-regression, in which autoregressive models are trained to reproduce one source and suppress the other. These linear approaches are tested on the problem of separating a known male from a known female speaker. The performance of these algorithms is assessed in terms of the residual error of estimated source spectrograms, waveform signal-to-noise ratio, and perceptual evaluation of speech quality scores. This work shows that the algorithms compare favorably to nonlinear approaches such as nonnegative sparse coding in terms of simplicity, performance, and suitability for real-time implementations, and they provide benchmark solutions for monaural source separation tasks.

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81. STICK: Spike Time Interval Computational Kernel, a Framework for General Purpose Computation Using Neurons, Precise Timing, Delays, and Synchrony

Xavier Lagorce, Ryad Benosman
https://doi.org/10.1162/neco_a_00783 | pp. 2261-2317

There has been significant research over the past two decades in developing new platforms for spiking neural computation. Current neural computers are primarily developed to mimic biology. They use neural networks, which can be trained to perform specific tasks to mainly solve pattern recognition problems. These machines can do more than simulate biology; they allow us to rethink our current paradigm of computation. The ultimate goal is to develop brain-inspired general purpose computation architectures that can breach the current bottleneck introduced by the von Neumann architecture. This work proposes a new framework for such a machine. We show that the use of neuron-like units with precise timing representation, synaptic diversity, and temporal delays allows us to set a complete, scalable compact computation framework. The framework provides both linear and nonlinear operations, allowing us to represent and solve any function. We show usability in solving real use cases from simple differential equations to sets of nonlinear differential equations leading to chaotic attractors.

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82. Visual Decisions in the Presence of Measurement and Stimulus Correlations

Manisha Bhardwaj, Samuel R. Carroll, Wei Ji, Krešimir Josi■■

https://doi.org/10.1162/neco_a_00778 | pp. 2318-2353

Humans and other animals base their decisions on noisy sensory input. Much work has been devoted to understanding the computations that underlie such decisions. The problem has been studied in a variety of tasks and with stimuli of differing complexity. However, how the statistical structure of stimuli, along with perceptual measurement noise, affects perceptual judgments is not well understood. Here we examine how correlations between the components of a stimulus-stimulus correlations-together with correlations in sensory noise, affect decision making. As an example, we consider the task of detecting the presence of a single or multiple targets among distractors. We assume that both the distractors and the observer's measurements of the stimuli are correlated. The computations of an optimal observer in this task are nontrivial yet can be analyzed and understood intuitively. We find that when distractors are strongly correlated, measurement correlations can have a strong impact on performance. When distractor correlations are weak, measurement correlations have little impact unless the number of stimuli is large. Correlations in neural responses to structured stimuli can therefore have a strong impact on perceptual judgments.

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83. Hierarchical Error Representation: A Computational Model of Anterior Cingulate and Dorsolateral Prefrontal Cortex

William H. Alexander, Joshua W. Brown

https://doi.org/10.1162/neco_a_00779 | pp. 2354-2410

Anterior cingulate and dorsolateral prefrontal cortex (ACC and dlPFC, respectively) are core components of the cognitive control network. Activation of these regions is routinely observed in tasks that involve monitoring the external environment and maintaining information in order to generate appropriate responses. Despite the ubiquity of studies reporting coactivation of these two regions, a consensus on how they interact to support cognitive control has yet to emerge. In this letter, we present a new hypothesis and computational model of ACC and dlPFC. The error representation hypothesis states that multidimensional error signals generated by ACC in response to surprising outcomes are used to train representations of expected error in dlPFC, which are then associated with relevant task stimuli. Error representations maintained in dlPFC are in turn used to modulate predictive activity in ACC in order to generate better estimates of the likely outcomes of actions. We formalize the error representation hypothesis in a new computational model based on our previous model of ACC. The hierarchical error representation (HER) model of ACC/dlPFC suggests a mechanism by which hierarchically organized layers within ACC and dlPFC interact in order to solve sophisticated cognitive tasks. In a series of simulations, we demonstrate the ability of the HER model to autonomously learn to perform structured tasks in a manner comparable to human performance, and we show that the HER model outperforms current deep learning networks by an order of magnitude.

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84. Bayesian Feature Selection with Strongly Regularizing Priors Maps to the Ising Model

Charles K. Fisher, Pankaj Mehta

https://doi.org/10.1162/neco_a_00780 | pp. 2411-2422

Identifying small subsets of features that are relevant for prediction and classification tasks is a central problem in machine learning and statistics. The feature selection task is especially important, and computationally difficult, for modern data sets where the number of features can be comparable to or even exceed the number of samples. Here, we show that feature selection with Bayesian inference takes a universal form and reduces to calculating the magnetizations of an Ising model under some mild conditions. Our results exploit the observation that the evidence takes a universal form for strongly regularizing priors--priors that have a large effect on the posterior probability even in the infinite data limit. We derive explicit expressions for feature selection for generalized linear models, a large class of statistical techniques that includes linear and logistic regression. We illustrate the power of our approach by analyzing feature selection in a logistic regression-based classifier trained to distinguish between the letters B and D in the notMNIST data set.

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85. A Novel Parameter Estimation Method for Boltzmann Machines

Takashi Takenouchi

https://doi.org/10.1162/neco_a_00781 | pp. 2423-2446

We propose a novel estimator for a specific class of probabilistic models on discrete spaces such as the Boltzmann machine. The proposed estimator is derived from minimization of a convex risk function and can be constructed without calculating the normalization constant, whose computational cost is exponential order. We investigate statistical properties of the proposed estimator such as consistency and asymptotic normality in the framework of the estimating function. Small experiments show that the proposed estimator can attain comparable performance to the maximum likelihood expectation at a much lower computational cost and is applicable to high-dimensional data.

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86. Bandit-Based Task Assignment for Heterogeneous Crowdsourcing

Hao Zhang, Yao Ma, Masashi Sugiyama

https://doi.org/10.1162/neco_a_00782 | pp. 2447-2475

We consider a task assignment problem in crowdsourcing, which is aimed at collecting as many reliable labels as possible within a limited budget. A challenge in this scenario is how to cope with the diversity of tasks and the task-dependent reliability of workers; for example, a worker may be good at recognizing the names of sports teams but not be familiar with cosmetics brands. We refer to this practical setting as heterogeneous crowdsourcing. In this letter, we propose a contextual bandit formulation for task assignment in heterogeneous crowdsourcing that is able to deal with the exploration-exploitation trade-off in worker selection. We also theoretically investigate the regret bounds for the proposed method and demonstrate its practical usefulness experimentally.

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87. Periodic Forcing of Inhibition-Stabilized Networks: Nonlinear Resonances and Phase-Amplitude Coupling

Romain Veltz, Terrence J. Sejnowski

https://doi.org/10.1162/neco_a_00786 | pp. 2477-2509

Inhibition-stabilized networks (ISNs) are neural architectures with strong positive feedback among pyramidal neurons balanced by strong negative feedback from inhibitory interneurons, a circuit element found in the hippocampus and the primary visual cortex. In their working regime, ISNs produce damped oscillations in the [Formula: see text]-range in response to inputs to the inhibitory population. In order to understand the properties of interconnected ISNs, we investigated periodic forcing of ISNs. We show that ISNs can be excited over a range of frequencies and derive properties of the resonance peaks. In particular, we studied the phase-locked solutions, the torus solutions, and the resonance peaks. Periodically forced ISNs respond with (possibly multistable) phase-locked activity, whereas networks with sustained intrinsic oscillations respond more dynamically to periodic inputs with tori. Hence, the dynamics are surprisingly rich, and phase effects alone do not adequately describe the network response. This strengthens the importance of phase-amplitude coupling as opposed to phase-phase coupling in providing multiple frequencies for multiplexing and routing information.

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88. Rhythmic Inhibition Allows Neural Networks to Search for Maximally Consistent States

Hesham Mostafa, Lorenz K. Müller, Giacomo Indiveri

https://doi.org/10.1162/neco_a_00785 | pp. 2510-2547

Gamma-band rhythmic inhibition is a ubiquitous phenomenon in neural circuits, yet its computational role remains elusive. We show that a model of gamma-band rhythmic inhibition allows networks of coupled cortical circuit motifs to search for network configurations that best reconcile external inputs with an internal consistency model encoded in the network connectivity. We show that Hebbian plasticity allows the networks to learn the consistency model by example. The search dynamics driven by rhythmic inhibition enable the described networks to solve difficult constraint satisfaction problems without making assumptions about the form of stochastic fluctuations in the network. We show that the search dynamics are well approximated by a stochastic sampling process. We use the described networks to reproduce perceptual multistability phenomena with switching times that are a good match to experimental data and show that they provide a general neural framework that can be used to model other perceptual inference phenomena.

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89. Learning Spatiotemporally Encoded Pattern Transformations in Structured Spiking Neural Networks

Gardner Brian, Ioana Sporea, André Grüning

https://doi.org/10.1162/neco_a_00790 | pp. 2548-2586

Information encoding in the nervous system is supported through the precise spike timings of neurons; however, an understanding of the underlying processes by which such representations are formed in the first place remains an open question. Here we examine how multilayered networks of spiking neurons can learn to encode for input patterns using a fully temporal coding scheme. To this end, we introduce a new

supervised learning rule, MultilayerSpiker, that can train spiking networks containing hidden layer neurons to perform transformations between spatiotemporal input and output spike patterns. The performance of the proposed learning rule is demonstrated in terms of the number of pattern mappings it can learn, the complexity of network structures it can be used on, and its classification accuracy when using multispike-based encodings. In particular, the learning rule displays robustness against input noise and can generalize well on an example data set. Our approach contributes to both a systematic understanding of how computations might take place in the nervous system and a learning rule that displays strong technical capability.

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90. A Sparse Reformulation of the Green's Function Formalism Allows Efficient Simulations of Morphological Neuron Models

Willem A. M. Wybo, Daniele Boccalini, Benjamin Torben-Nielsen, Marc-Oliver Gewaltig

https://doi.org/10.1162/neco_a_00788 | pp. 2587-2622

We prove that when a class of partial differential equations, generalized from the cable equation, is defined on tree graphs and the inputs are restricted to a spatially discrete, well chosen set of points, the Green's function (GF) formalism can be rewritten to scale as O(n) with the number n of inputs locations, contrary to the previously reported O(n(2)) scaling. We show that the linear scaling can be combined with an expansion of the remaining kernels as sums of exponentials to allow efficient simulations of equations from the aforementioned class. We furthermore validate this simulation paradigm on models of nerve cells and explore its relation with more traditional finite difference approaches. Situations in which a gain in computational performance is expected are discussed.

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91. Positive Neural Networks in Discrete Time Implement Monotone-Regular Behaviors

Tom J. Ameloot, Jan Van den Bussche

https://doi.org/10.1162/neco_a_00789 | pp. 2623-2660

We study the expressive power of positive neural networks. The model uses positive connection weights and multiple input neurons. Different behaviors can be expressed by varying the connection weights. We show that in discrete time and in the absence of noise, the class of positive neural networks captures the so-called monotone-regular behaviors, which are based on regular languages. A finer picture emerges if one takes into account the delay by which a monotone-regular behavior is implemented. Each monotone-regular behavior can be implemented by a positive neural network with a delay of one time unit. Some monotone-regular behaviors can be implemented with zero delay. And, interestingly, some simple monotone-regular behaviors cannot be implemented with zero delay.

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92. Symbolic Computation Using Cellular Automata-Based Hyperdimensional Computing

Özgür Y■lmaz

https://doi.org/10.1162/neco_a_00787 | pp. 2661-2692

This letter introduces a novel framework of reservoir computing that is capable of both connectionist machine intelligence and symbolic computation. A cellular automaton is used as the reservoir of dynamical systems. Input is randomly projected onto the initial conditions of automaton cells, and nonlinear computation is performed on the input via application of a rule in the automaton for a period of time. The evolution of the automaton creates a space-time volume of the automaton state space, and it is used as the reservoir. The proposed framework is shown to be capable of long-term memory, and it requires orders of magnitude less computation compared to echo state networks. As the focus of the letter, we suggest that binary reservoir feature vectors can be combined using Boolean operations as in hyperdimensional computing, paving a direct way for concept building and symbolic processing. To demonstrate the capability of the proposed system, we make analogies directly on image data by asking, What is the automobile of air?